



Asset Performance:

Ene 16 Monitoring tenanted areas



Credits



No Minimum Standard

Aim

To reduce operational energy consumption through the effective management and monitoring of energy consumption in separately tenanted and functional areas.

Question

Is energy consumption for separately tenanted areas sub-metered?

Credits	Answer	Select all that apply
0	A.	Question not answered
1	B.	All separately tenanted areas with significant energy consumption are sub-metered.
1	C.	All end uses with significant energy consumption within each separately tenanted area are sub-metered.

Is the energy consumption for functional areas sub-metered?

Credits	Answer	Select all that apply
0	D.	Question not answered
1	E.	Functional areas with significant energy consumption are sub-metered.
1	F.	All end uses with significant energy consumption within each functional area are sub-metered.

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering Where the asset does not have any tenanted areas, or where the estimated energy consumption for the tenanted areas is not significant, this issue can be filtered out of the assessment. Where the asset does not have any functional areas with significantly different energy use patterns, or where the estimated energy consumption	A - C D - F

Criterion	Assessment criteria	Applicable Answer
	for the functional area is not significant this issue can be filtered out of the assessment.	
2.	Significant energy consumption is considered to be >8,500kWh per year for electricity and >67,000kWh per year for other energy sources.	A - F
3.	Functional areas must be separately sub-metered where the energy use patterns are expected to be significantly different from that of the main asset	D - F
4.	Where there is more than one HVAC system type serving a particular end use the requirement applies to the main HVAC system.	C , F
5.	Where more than one end use is provided by the same HVAC component it is acceptable to sub-meter the combined energy consumption.	C , F

Methodology

Sub-metering separately tenanted areas:

For each fuel type used estimate the energy consumption for each separately tenanted area to determine whether it is significant.

In the absence of more accurate information, energy consumption in tenanted areas may be estimated based on total energy consumption per m² for the asset and the floor area of the tenanted areas

Sub-metering functional areas with significantly different energy use patterns:

Identify functional areas within the asset which have a significantly different energy use pattern from that of the rest of the asset.

A functional area is considered to be significantly different where the occupancy hours or the density and use of installed services and equipment are likely to be significantly different from that of the main asset.

Examples of functional areas that are likely to have significantly different energy use patterns includes; commercial scale kitchens, sports and leisure facilities, conference suites, server rooms and cold storage areas.

Sub-metering end use within separately tenanted and functional areas with significantly different energy use patterns:

Follow the methodology in Ene 15 Monitoring energy uses.

Sub-metering by calculation:

It is acceptable to calculate the energy consumption for an end-use by subtracting sub-metered energy consumption for other end-uses from the relevant main utility meter reading.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Estimated energy consumption for tenanted areas, functional areas with different energy requirements and high energy intensity equipment and areas.
All	Either copies of verified sub-meter data for the first and last date of the 12-month period specified. These may be outputs from energy monitoring and management systems or automatic or manual meter readings OR; Line diagram indicating sub-meters and the related energy uses or evidence to show that the tenanted area can be monitored separately.

Definitions

Different functional areas:

A functional area is different where the occupancy hours or the internal environmental conditions (e.g. temperature, ventilation rate) differ significantly from those that apply to the rest of the asset.

Energy monitoring and management system:

Examples include automatic meter reading systems and building energy management systems (BEMS). Automatic monitoring and targeting is an example of a management tool that includes automatic meter reading and data management.

Small plug in loads:

Plug-in equipment/appliances connected through power points.

Significant energy consumption:

An end use or building servicing system is considered to be significant where the typical energy cost savings achieved through sub-metering is expected to payback within 10 years through energy cost savings achieved through improved energy management. For the purpose of this issue, energy use is where the estimated energy consumption exceeds the threshold kWh/year values of 8,500 kWh/year for electricity and 67,000 kWh/year for other fuels are deemed to be significant.

Sub-meters:

Sub-meters are secondary to the main utility meters and are installed to measure consumption by specific items of plant or equipment, or to discrete physical areas, e.g. individual buildings, floors in a multi-storey building, tenanted areas, functional areas. Outputs include pulsed outputs or other open protocol communication outputs.